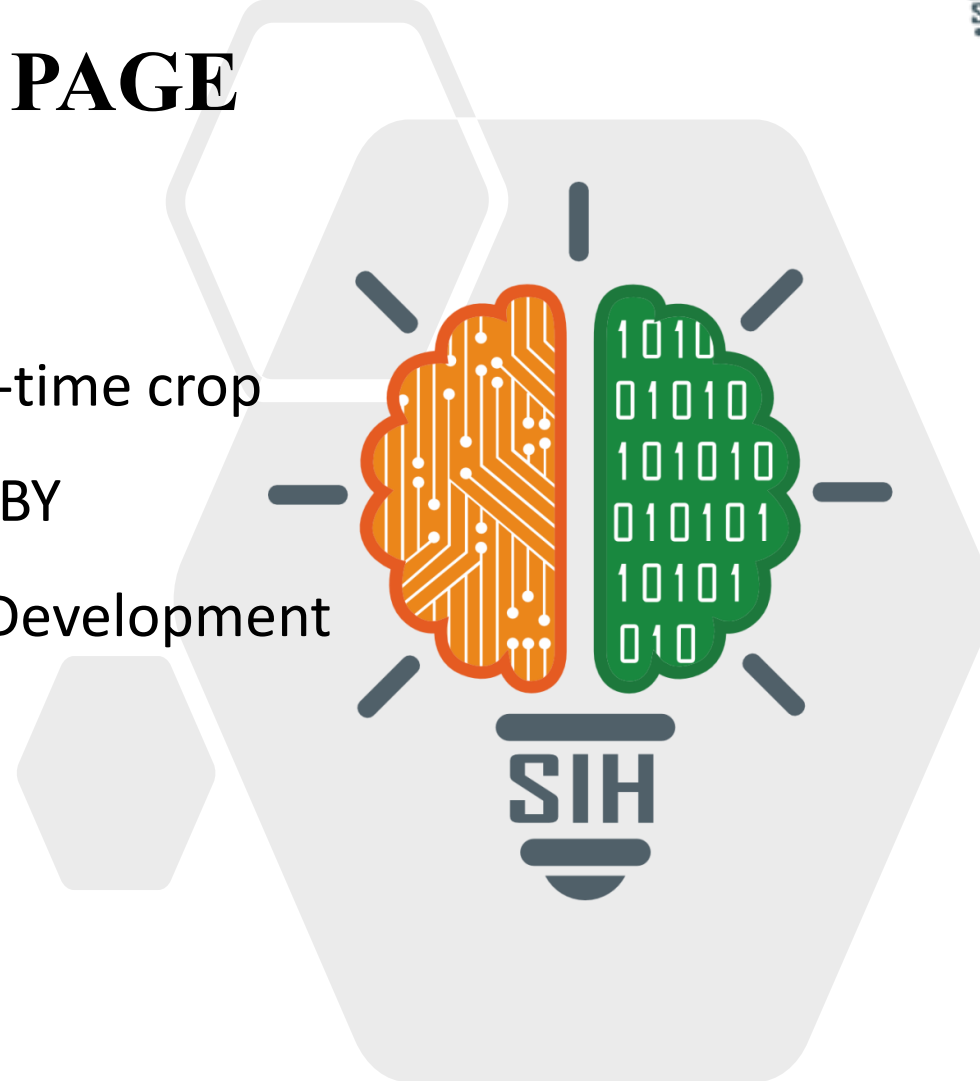


SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID** - SIH25262
- **Problem Statement Title**- AI based real-time crop image analytics for crop insurance-PMFBY
- **Theme**- Agriculture, FoodTech & Rural Development
- **PS Category**- Software
- **Team ID**- 74859
- **Team Name**- NOVA@SBU



AI Smart Crop Image Analytics for Insurance



❖ Proposed Solution:

- ❑ A real-time AI solution that uses geo-tagged crop images and deep learning to assess crop health, growth, and damage, ensuring faster and unbiased insurance claim processing.

❖ Mobile Application (Farmer & Field Officer Use):

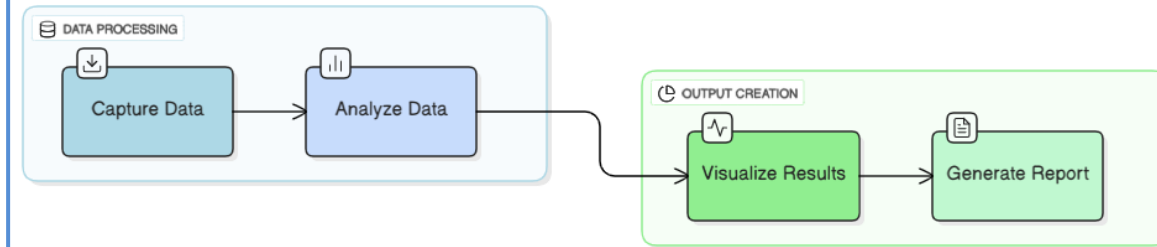
- ❑ Capture and upload **geo-tagged crop images** at key growth stages.
- ❑ Built-in **guidance system** for correct image framing and crop selection.
- ❑ Offline storage & sync support for rural areas.

❖ AI/ML Cloud Platform:

- ❑ **Image validation:** Checks for quality, relevance, and duplicates.
- ❑ **AI Analysis:** Uses deep learning (CNN models) for:
 - Crop classification
 - Growth stage detection
 - Damage detection (pest, flood, drought, etc.)

❖ Web Dashboard (For Government/Insurance Officials):

- ❑ Real-time **map-based visualization** of crop data.
- ❑ Analytics and alerts for reported damages.
- ❑ Downloadable claim verification reports.



❖ Innovation and Uniqueness

- ❑ **AI + Mobile + Cloud integrated system** for agriculture insurance.
- ❑ **End-to-end digital workflow** (capture → analyze → visualize → report).
- ❑ **AI model trained for Indian crop diversity** and damage patterns.
- ❑ **Offline-first mobile app** for remote regions.
- ❑ Compatible with **future PMFBY and YESTECH frameworks**.

❖ Technologies to be Used:

- ❑ **Frontend:** Flutter (for mobile app), React.js (dashboard), TypeScript, TailwindCSS, Framer Motion, React Router
- ❑ **Backend:** Node.js, Express.js
- ❑ **Database:** MongoDB / Firebase
- ❑ **AI/ML Models:** Python, TensorFlow / PyTorch, OpenCV
- ❑ **Cloud Services:** Google Cloud (for data storage & model hosting, For APIs)
- ❑ **APIs:** Geo-tagging APIs, Weather API integration, REST API
- ❑ **Maps, Geo, Notifications, Report Generation:** Mapbox / Leaflet.js with GeoJSON, FCM or EmailJS, jsPDF / PDFKit

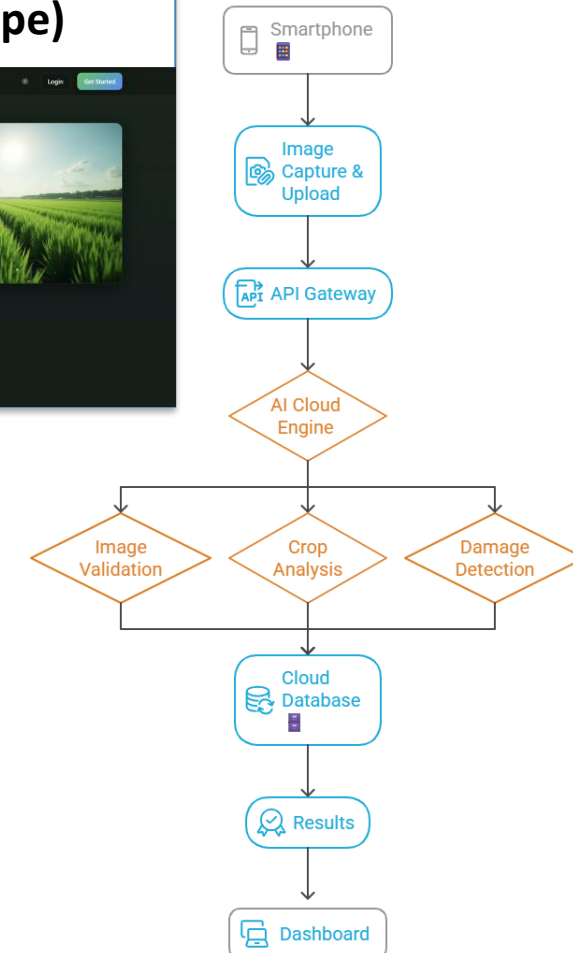
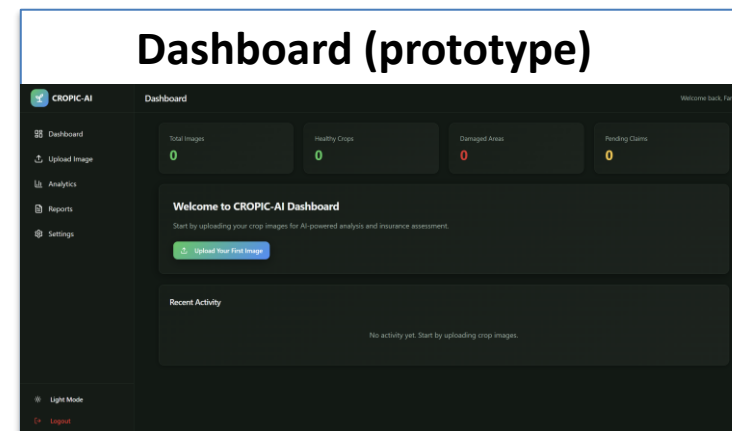
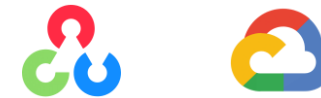
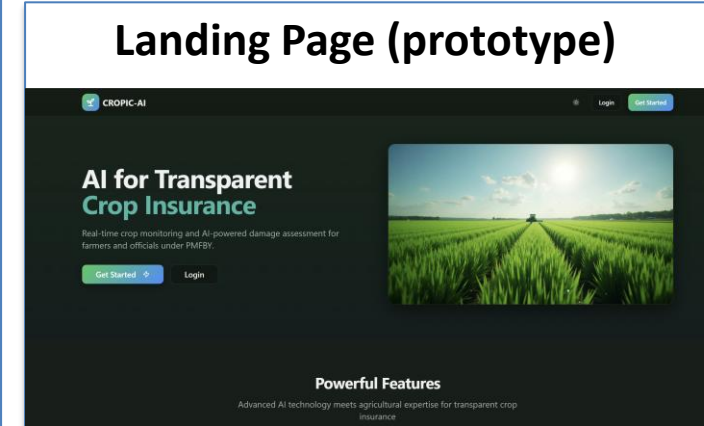
❖ Methodology & Process Flow:

❑ System Workflow:

1. Farmer captures and uploads crop images (with location, time, crop stage).
2. Images go to **AI Cloud Engine** for:
 - I. Image validation
 - II. Crop and damage analysis
3. Results are stored in database and visualized on dashboard.
4. Officials review analytics and verify claims digitally.



Smart Agriculture Solution Architecture



❖ Feasibility Analysis:

❑ Technical Feasibility:

- Built using robust, **open-source technologies** (React, Node.js, TensorFlow, MongoDB).
- Cloud-based and modular — easy to scale nationwide under PMFBY.
- AI models can be continuously trained using real-world datasets for improved accuracy.

❑ Operational Feasibility:

- Simple UI/UX for farmers and officials, accessible via mobile or low-end devices.
- Geo-tagged image uploads ensure authenticity and traceability.
- Works seamlessly with existing government portals and YESTECH systems.

❑ Economic Feasibility:

- Low infrastructure cost using cloud hosting (AWS/Firebase).
- Reduces manual survey and labor costs significantly.
- Long-term savings through automation and transparency.

❖ Potential Challenges & Risks:

❑ Diverse crop and lighting conditions:

- Different soil, crop, and weather variations may affect AI accuracy

❑ Limited rural internet access:

- Farmers in remote areas may face upload delays or connectivity issues.

❑ Data privacy concerns:

- Handling of farmer data and geolocation needs strong security measures.

❑ Model drift over time:

- AI performance may degrade without periodic retraining.

Challenge	Solution Strategy
Diverse crop conditions	Use large, region-specific datasets and augment images during training.
Poor connectivity	Implement offline data caching and auto-sync once network is available.
Data privacy	Enable end-to-end encryption, secure APIs, and role-based authentication.
Model drift	Set up a periodic retraining pipeline using newly collected real-time data.

❖ Potential Impact:

❑ Transforms PMFBY Operations:

- Brings end-to-end digital automation in crop insurance processing through AI.

❑ Faster Claim Settlements:

- Real-time damage detection and report generation reduce settlement time from months to days.

❑ Transparency & Trust:

- AI and geotagged validation build accountability among farmers, insurers, and officials.

❑ Nationwide Crop Health Insights:

- Enables data-driven decision-making through continuous crop analytics and satellite monitoring.

❑ Climate Resilience Enhancement:

- Early alerts for droughts, pests, or floods strengthen disaster response and policy planning.

❑ Empowers Rural Farmers:

- Makes high-end agri-tech accessible to smallholders, bridging the digital divide across India.

Key benefits



Social Benefits

Empowers farmers and promotes digital inclusion



Economic Benefits

Reduces costs and boosts financial resilience



Environmental Benefits

Enables early detection of crop stress and promotes sustainability



Technological Benefits

Provides a scalable AI-cloud ecosystem

❖ References:

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👉 <https://agricoop.gov.in>
- ❑ *AI in Agriculture: Precision Farming and Crop Monitoring* – International Publications Journal, 2024
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- ❑ *Precision Agriculture for Improving Crop Yield Predictions* – Frontiers in Agronomy, 2025
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👉 <https://pmc.ncbi.nlm.nih.gov/articles/PMC11751057/>

❖ Dataset References:

- ❑ Indian Crop Yield Dataset (ICID) – State-wise agricultural and climatic data
👉 <https://gts.ai/dataset-download/indian-crop-yield-dataset/>
- ❑ PlantVillage Dataset (Open Source) – Global standard for leaf disease identification
👉 <https://www.kaggle.com/datasets/kamal01/top-agriculture-crop-disease>
- ❑ PlantDoc Dataset – Visual dataset for Indian crop disease detection
👉 <https://github.com/pratikkayal/PlantDoc-Dataset>
- ❑ MH-Weed16 Dataset – Annotated Indian weed and crop dataset for AI tasks
👉 <https://data.mendeley.com/datasets/d3n3mgjibv/1>
- ❑ SoyNet Dataset – Indian soybean disease image dataset
👉 <https://data.mendeley.com/datasets/w2r855hpx8/2>
- ❑ AgriFieldNet India Challenge Dataset – Sentinel-2 satellite crop classification data
👉 <https://zindi.africa/competitions/agrifieldnet-india-challenge>